

TWO HARBORS HELGESON, MN, USA

WMO: 727444

Lat: 47.049N

Lon: 91.745W

Elev: 1080

StdP: 14.13

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04979

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.1	-10.9	-26.5	1.3	-16.0	-20.5	1.9	-8.7	20.1	22.4	18.4	20.7	5.3	290	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	83.8	71.8	81.3	68.9	78.8	66.6	73.8	81.5	71.0	78.9	68.4	75.8	8.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.7	122.0	80.1	68.1	107.6	76.3	65.7	98.7	73.9	38.1	81.4	35.4	78.8	33.3	76.0	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.2	17.6	15.4	DB	-23.1	90.0	6.7	3.2	-28.0	92.3	-31.9	94.2	-35.7	96.0	-40.5	98.3	(4)
(5)			WB	-23.5	77.1	6.3	3.1	-28.0	79.4	-31.7	81.2	-35.3	83.0	-39.9	85.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.0	11.8	15.2	26.5	38.5	50.1	59.5	65.8	63.7	55.8	43.6	30.0	17.7		
	DBStd	20.89	13.14	12.16	11.69	8.53	7.87	6.62	5.93	5.70	7.81	7.73	10.46	11.83		
	HDD50	5219	1186	976	730	355	96	8	0	0	33	231	601	1003		
	HDD65	9313	1651	1396	1194	794	466	187	63	92	289	664	1049	1468		
	CDD50	1559	0	0	2	11	99	291	491	425	206	32	2	0		
	CDD65	178	0	0	0	0	4	20	89	52	13	0	0	0		
	CDH74	1741	0	0	0	3	101	258	816	448	111	4	0	0		
	CDH80	364	0	0	0	0	20	44	194	92	14	0	0	0		
	WSAvg	6.0	6.2	6.3	6.4	7.3	6.9	5.7	5.1	5.0	5.3	6.3	6.2	6.0		
	PrecAvg	30.9	1.1	1.1	1.5	2.6	3.1	3.8	3.8	3.3	3.7	3.3	2.2	1.4		
(15) Precipitation	PrecMax	37.7	2.9	3.4	3.4	8.4	5.2	6.5	9.5	6.5	6.9	6.8	6.7	2.5		
	PrecMin	24.2	0.2	0.3	0.4	0.6	0.9	0.5	1.1	1.0	0.4	1.1	0.3	0.2		
	PrecStd	4.1	0.7	0.7	0.7	1.6	1.1	1.5	2.1	1.4	1.6	1.4	1.5	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.2	46.2	60.8	72.0	82.2	84.8	88.8	87.9	81.8	72.3	62.7	44.7		
		MCWB	37.4	39.9	51.1	55.2	64.6	73.3	74.3	75.1	69.6	59.8	52.6	39.4		
	2%	DB	36.7	39.4	52.3	63.8	75.4	80.8	84.2	81.9	77.2	65.9	54.0	38.9		
		MCWB	33.7	35.2	44.0	50.9	60.2	66.7	72.2	70.7	67.0	57.5	49.1	36.2		
	5%	DB	33.8	36.3	46.0	58.7	70.6	76.8	81.7	79.2	72.9	61.6	48.0	36.4		
		MCWB	31.4	33.6	40.3	47.2	57.4	64.3	69.9	67.7	64.3	53.7	43.4	34.0		
	10%	DB	29.8	32.5	42.5	53.9	64.4	73.0	79.2	75.2	70.0	56.7	44.9	34.0		
		MCWB	28.2	30.3	37.2	44.4	54.4	62.6	67.5	65.7	62.2	50.6	40.4	31.7		
	0.4%	WB	37.9	41.2	51.8	58.0	67.5	75.2	77.0	76.5	73.3	62.9	54.4	40.1		
		MCDB	42.8	45.9	57.7	66.6	79.7	82.6	84.4	85.5	79.8	67.6	60.6	43.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.8	36.6	44.9	52.1	62.4	69.1	74.4	72.6	68.7	58.9	48.7	36.8		
		MCDB	36.5	39.6	49.5	62.6	72.0	77.2	82.2	80.6	75.3	64.7	53.5	39.1		
	5%	WB	31.3	33.4	40.7	48.1	59.0	66.4	71.8	69.6	65.9	55.0	44.3	33.4		
		MCDB	33.4	35.9	46.0	57.9	68.2	74.2	80.1	76.6	71.9	60.5	47.6	35.9		
	10%	WB	27.9	30.2	37.3	44.7	55.7	63.9	69.0	67.3	63.0	51.1	40.3	31.7		
		MCDB	29.1	32.4	42.2	52.5	64.1	70.9	76.7	74.2	67.9	55.7	44.3	33.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.8	18.4	17.6	18.2	21.0	20.6	20.6	19.7	19.1	16.4	14.3	14.4		
		MCDBR	16.6	19.8	22.5	29.3	30.5	26.6	24.2	23.8	24.2	25.1	20.8	14.7		
		MCWBR	14.3	15.8	16.2	18.8	17.8	16.1	14.9	14.8	16.3	17.2	15.4	12.3		
	5% WB	MCDBR	15.8	17.3	21.5	27.7	26.8	23.7	23.2	21.7	21.7	22.5	19.0	13.7		
		MCWBR	13.9	14.1	15.7	18.6	17.3	16.1	15.5	15.0	15.7	16.6	14.8	11.8		
(40) Clear-Sky Solar Irradiance	taub		0.249	0.278	0.314	0.344	0.372	0.377	0.385	0.382	0.361	0.313	0.288	0.266		
	taud		2.424	2.352	2.359	2.384	2.369	2.386	2.390	2.421	2.450	2.560	2.521	2.431		
	Ebn at Noon		274	285	289	289	283	281	277	274	270	271	257	252		
	Edh at Noon		21	28	33	35	37	37	36	34	30	23	19	19		
(44) All-Sky Solar Radiation	RadAvg		423	749	1130	1459	1697	1830	1933	1623	1213	733	419	308		
	RadStd		46	73	104	150	123	128	79	111	77	88	52	31		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page